

MATTHEW YU

(226) · 606 · 9496 ◇ mnyu@uwaterloo.ca ◇

EDUCATION

University of Waterloo

Bachelor of Applied Science (B.A.Sc.) in Computer Engineering

September 2025 - April 2030

Waterloo, ON

EXPERIENCE

HavenIQ — Software Engineer

January 2026 - April 2026

- Built real-time IoT data pipelines on **AWS** using Lambda, ECS, and EventBridge to ingest and process telemetry from connected devices into a **BigQuery** database
- Developed and evaluated **PyTorch** neural networks and **XGBoost** models on large-scale sensor telemetry for risk analysis, achieving strong predictive performance ($R^2 = 0.95$) on validation data
- Designed cloud analytics infrastructure using AWS Lambda, ECS Fargate, EventBridge, CloudWatch, and Secrets Manager to process telemetry and power real-time monitoring
- Developed integrations for third-party APIs using **OAuth 2.0**, enabling reliable ingestion of data from multiple device and weather platforms
- Deployed containerized **MQTT** ingestion services on AWS ECS Fargate using Docker, CloudWatch, and DevOps practices for reliable event streaming

PROJECTS

HavenIQ HVAC RAG System | *Python, FastAPI, PyTorch, Qdrant, Docker, OpenAI API*
May 2026

- Built a **FastAPI**-based **RAG** system that uses sensor embeddings and retrieves historical incidents to explain sensor data anomalies using **OpenAI LLMs**
- Developed a telemetry pipeline that converts raw HVAC readings into sliding-window feature tensors and trains a custom 128-dimensional **PyTorch** encoder
- Integrated **Qdrant Cloud** vector search with metadata filtering for similarity retrieval over historical HVAC incidents

Agentic Ecommerce Platform | *Python, LangGraph, Supabase, Stripe, APIs* April 2026

- Built a **\$5K+** revenue e-commerce platform with **Next.js**, **Stripe**, **Supabase**, and marketplace APIs
- Developed a **LangGraph** agent for order ingestion, SKU mapping, and fulfillment orchestration
- Integrated **LLM tool-calling** workflows for supplier routing, exceptions, and human-in-the-loop review
- Implemented guardrails and audit logs to ensure reliable, traceable order processing

AlphaZero-Style Othello Agent | *Python, NumPy, PyTorch, Matplotlib* March 2026

- Built, tested, and refined a deep reinforcement learning agent to play Othello using **self-play**, **MCTS**, and a residual convolutional policy-value network in **PyTorch**
- Optimized Python ML prototype performance with **Numba @njit** JIT compilation and multiprocessing CPU workers with **CUDA** inference, increasing self-play data generation speed by over **60×**
- Conducted model evaluation, benchmarking, and exploratory analysis with custom training metrics logged to CSV and visualized in **Matplotlib**, enabling targeted rollbacks instead of full restarts

TECHNICAL SKILLS

Languages	Python, C++, SQL
Frameworks	FastAPI, PyTorch, XGBoost, scikit-learn, NumPy, pandas
Infrastructure	AWS (Lambda, ECS, S3, CloudWatch, EventBridge), Docker, BigQuery
Tools	MS Excel, Git/GitHub, Postman, Airtable, REST APIs, Vercel